

Curriculum Vitae – Niels Slotboom

MASt Theoretical Physics / PhD Student

Location: Cambridge, UK

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Academic Summary

I am a theoretical physicist working at the intersection of mathematical gravitation, numerical relativity, and high-performance computing. My academic path has been driven by a deep appreciation for the geometric rigour of spacetime physics and the computational frameworks needed to simulate it. My research focuses on the entire pipeline of strong-field gravitational problems, with a particular emphasis on designing versatile parabolic/diffusion-type solvers for initial data via functional gradient descent and exploring the non-linear classical evolution regimes of higher-order curvature frameworks like Quadratic Gravity. This winter, I am beginning my PhD at the University of Nottingham to continue developing numerical methods for these extended theories of gravity.

Education

2026 - 2030	School of Mathematics, University of Nottingham , Nottingham, UK <i>PhD Mathematics</i> Supervisors: Miguel Bezares, Thomas Sotiriou Area: Numerical Relativity in Modified Gravity
2025 - 2026	DAMTP, University of Cambridge , Cambridge, UK <i>Part III, MASt in Theoretical Physics</i> Grade: Distinction (91/100) Ranking: 8th out of 284 Honours: Awarded the Mayhew Prize for greatest distinction in Applied Mathematics Essay: "General Relativity as a Dynamical PDE"
2023 - 2025	Institute of Theoretical Physics, University of Bern , Bern, CH <i>MSc in Theoretical Physics</i> Grade: 5.96 / 6.0 (99.2%), Thesis: Recovering the Eikonal Quasinormal Mode Spectrum of the Schwarzschild Black Hole via the Penrose Limit
2019 - 2023	Department of Physics, University of Bern , Bern, CH <i>BSc in Physics, Minors in Mathematics and Computer Science</i> Grade: 5.74 / 6.0 (94.8%), Thesis: Development and Theoretical Study of RF Pulses for Globally Spatially Homogeneous Excitation in Magnetic Resonance Imaging at (X)UHF
2016 - 2019	Gymnasium Lerbermatt , Köniz, CH <i>Swiss Matura</i>

Professional Experience

7/2026 - 11/2026	Bau- und Verkehrsdirektion Kanton Bern <i>Systems & Software Engineer (Temporary Full-time Contract)</i> Transitioned to a full employee role to lead the production deployment, optimisation, and scaling of the LiDAR-based embedded vehicle detection system developed during the 2025 internship. Focus on refining real-time processing performance and hardening the system architecture for active traffic light control integration.
4/2025 - 9/2025	Bau- und Verkehrsdirektion Kanton Bern <i>Internship on Traffic Management Systems</i> Lead development of a LiDAR-based embedded system for vehicle detection and counting, designed for traffic light control at intersections and traffic surveys. Further, I automated data analysis processes in C++ to improve performance evaluations and reporting.
2/2024 - 2/2025	Institute of Theoretical Physics, University of Bern <i>Position as Teaching Assistant</i> Supervision of tutorial sessions, grading exercises and exams, and answering individual questions from bachelor's students for lectures on mathematical methods in physics.

Theses

Part III Essay (*Supervisor*: Rita Teixeira da Costa)

Title: General Relativity as a Dynamical PDE

Description: This essay synthesised the 1969 Choquet-Bruhat and Geroch theorem establishing the global well-posedness of the Einstein field equations. The work developed the ADM decomposition to derive the initial value constraints, used the Bochner formula under harmonic gauge to reduce the vacuum Ricci tensor to a quasilinear hyperbolic system, and applied Zorn's lemma to construct a unique, maximal globally hyperbolic development from the initial data. The project built a rigorous mathematical bridge between differential geometry and non-linear wave equations, directly aligning with my trajectory into numerical relativity. For more details, see [here](#).

Master Thesis (*Supervisor*: Matthias Blau)

Title: Recovering the Eikonal Quasinormal Mode Spectrum of the Schwarzschild Black Hole via the Penrose Limit

Description: This thesis investigated the eikonal quasinormal mode (QNM) spectrum of the Schwarzschild black hole, exploring how it emerges from the Penrose limit of the spacetime along an equatorial photon ring geodesic. The work involved three analytical approaches—WKB analysis, geometric optics, and wave propagation in the Penrose limit—all of which reproduced the known QNM spectrum. A primary result was identifying how local symmetries near the photon sphere become exact in the Penrose limit, explaining the agreement. The project deepened my expertise in both general relativity and differential geometry. For more details, see [here](#).

Bachelor Thesis (*Supervisor*: Thomas Feurer)

Title: Development and Theoretical Study of RF Pulses for Globally Spatially Homogeneous Excitation in Magnetic Resonance Imaging at (X)UHF

Description: This thesis focused on improving parallel transmission (pTx) MRI at high field strengths, particularly at 7T, by addressing RF interference-induced excitation inhomogeneity. I developed a simulation framework based on Maxwell's equations and quaternionic Bloch's equations, implemented in CUDA C to run on the GPU. This simulation framework was embedded in a gradient descent algorithm to iteratively optimise RF pulse parameters, effectively reducing excitation inhomogeneity compared to standard circular polarisation pulses. The work demonstrates a practical approach for real-time, patient-specific pulse optimisation, and is the basis of a still ongoing research project.

Skills

Relevant Coursework

- **Part III Courses**: General Relativity; Gravitational Waves and Numerical Relativity; Black Holes; Canonical Gravity; Differential Geometry; Analysis of PDEs; Symmetries, Particles and Fields; Solitons, Instantons and Geometry; Essay
- **Master's Courses**: General Relativity; Quantum Field Theory; Standard Model; Advanced Concepts in Theoretical Physics; Effective Field Theories; Theoretical Exercises; Thesis
- **Bachelor's Courses**: Classical Field Theories; Quantum Theory I–III; Statistical Thermodynamics I & II; Electrodynamics; Solid State Physics; Experimentalist Courses; Thesis

Further Courses in Mathematics

- **Mathematics**: Topology; Functional Analysis; ODEs; Non-linear PDEs; Algebra; Complex Analysis; Fourier Analysis; Real Analysis; Linear Algebra
- **Computer Science**: Programming (Java); Computer Architecture; Fundamentals of Technical Computer Science

Programming and Web

- **C++ / CUDA / C**: Advanced, high proficiency in HPC and numerics, approaching senior-level design and performance (multiple >10k LOC maintainable and extendible codebases)
- **Java / Python**: Moderate; used for scripting, tooling, and coursework
- **HTML / CSS**: Basic; sufficient for website authoring and layout

- **x86 Assembly / PTX:** Read-level proficiency; occasionally used for performance analysis and low-level optimisation insight

Technical Skills

- **Typesetting:** LaTeX, Typst
- **Hardware:** Experienced in Computer Configuration and Assembly, Hardware Architecture, Embedded Systems
- **Electronics:** Circuit and PCB Design, Repairs, Reverse Engineering
- **General:** Troubleshooting, Networking

Complementary Skills

- **Teaching:** Experienced in mentoring peers and serving as a teaching assistant for a full year, with a focus on adapting explanations to individual learning needs

Languages

- **German, Dutch:** Native
- **English:** C2 CEFR, Score of 209 on Cambridge Scale, acquired in 2024
- **French:** Baccalaureate level